INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Basic Computer Skills for Digital Forensics

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

A practical foundation for interpreting digital evidence at byte, text, time and memory levels

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Course: CIP-B101 - Basic Computer Skills for Digital Forensics

Lab Title: Lab 1B-PC System Fundamentals, Disk Sectors and Partition Analysis

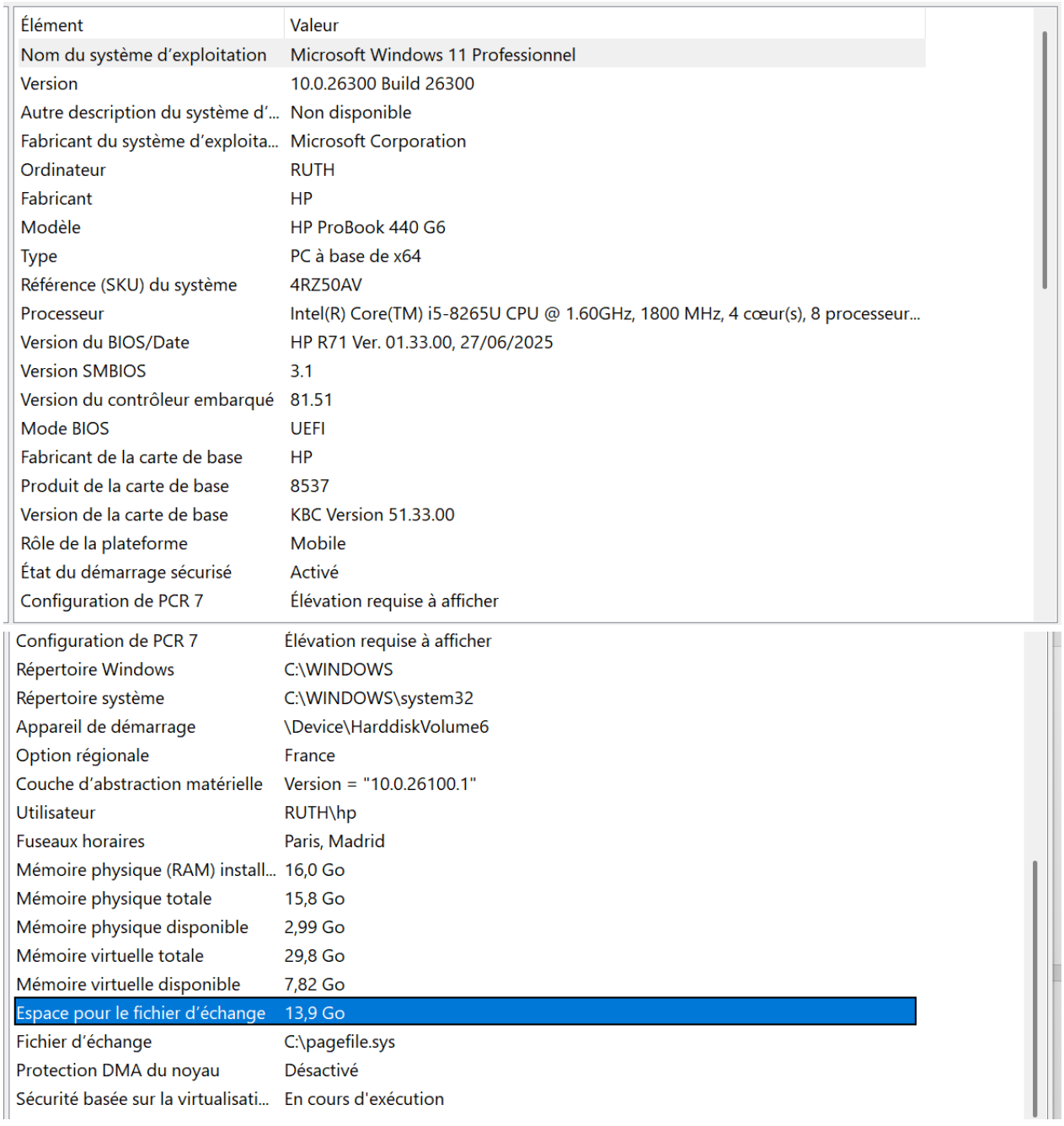
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Part A - System Identification and Evidence Layer Mapping

A1. Capture basic system specification

• Capture a screenshot showing OS Name, Version, System Type, Processor, BIOS Mode, Installed Physical Memory, and Secure Boot State if available.



Élément	Valeur
Nom du système d'exploitation	Microsoft Windows 11 Professionnel
Version	10.0.26300 Build 26300
Autre description du système d'...	Non disponible
Fabricant du système d'exploita...	Microsoft Corporation
Ordinateur	RUTH
Fabricant	HP
Modèle	HP ProBook 440 G6
Type	PC à base de x64
Référence (SKU) du système	4RZ50AV
Processeur	Intel(R) Core(TM) i5-8265U CPU @ 1.60GHz, 1800 MHz, 4 cœur(s), 8 processeur...
Version du BIOS/Date	HP R71 Ver. 01.33.00, 27/06/2025
Version SMBIOS	3.1
Version du contrôleur embarqué	81.51
Mode BIOS	UEFI
Fabricant de la carte de base	HP
Produit de la carte de base	8537
Version de la carte de base	KBC Version 51.33.00
Rôle de la plateforme	Mobile
État du démarrage sécurisé	Activé
Configuration de PCR 7	Élévation requise à afficher
Configuration de PCR 7	Élévation requise à afficher
Répertoire Windows	C:\WINDOWS
Répertoire système	C:\WINDOWS\system32
Appareil de démarrage	\Device\HarddiskVolume6
Option régionale	France
Couche d'abstraction matérielle	Version = "10.0.26100.1"
Utilisateur	RUTH\hp
Fuseaux horaires	Paris, Madrid
Mémoire physique (RAM) install...	16,0 Go
Mémoire physique totale	15,8 Go
Mémoire physique disponible	2,99 Go
Mémoire virtuelle totale	29,8 Go
Mémoire virtuelle disponible	7,82 Go
Espace pour le fichier d'échange	13,9 Go
Fichier d'échange	C:\pagefile.sys
Protection DMA du noyau	Désactivé
Sécurité basée sur la virtualisati...	En cours d'exécution

• Also run the command below in Command Prompt and capture the first page of output.

```
C:\Users\hp>systeminfo

Nom d'hôte:                                 RUTH
Nom du système d'exploitation:              Microsoft Windows 11 Professionnel
Version du système d'exploitation:          10.0.26300 N/A build 26300
Fabricant du système d'exploitation:        Microsoft Corporation
Configuration du système d'exploitation:    Station de travail membre
Type de build du système d'exploitation:    Multiprocessor Free
Propriétaire enregistré:                    hp
Organisation enregistrée:
ID du produit:                              00330-51988-79333-AAOEM
Date d'installation originale:              10/05/2025, 04:35:10
Heure de démarrage du système:              04/11/2026, 21:39:50
Fabricant du système:                       HP
Modèle du système:                          HP ProBook 440 G6
Type du système:                            x64-based PC
Processeur(s):                              1 processeur(s) installé(s).
                                            [01] : Intel64 Family 6 Model 142 Stepping 11 GenuineIntel ~1600 M
Version du BIOS:                            HP R71 Ver. 01.33.00, 27/06/2025
Répertoire Windows:                         C:\WINDOWS
Répertoire système:                         C:\WINDOWS\system32
Appareil de démarrage:                      \Device\HarddiskVolume6
Paramètres régionaux système:               fr;Français (France)
Paramètres régionaux d'entrée:              fr;Français (France)
Fuseau horaire:                             (UTC+01:00) Bruxelles, Copenhague, Madrid, Paris
Mémoire physique totale:                    16 223 Mo
Mémoire physique disponible:                2 976 Mo
Mémoire virtuelle : taille maximale:        30 496 Mo
```

A2. Capture storage information

```
PS C:\Users\hp> Get-Disk

Number Friendly Name Serial Number                    HealthStatus         OperationalStatus      Total Size Partition
                                                                                                           Style
------ ------------- -------------                    ------------         -----------------      ---------- ----------
0      HGST HTS72...       RCF50ACE0YUDTM             Healthy              Online                  465.76 GB GPT
1      WDC PC SN5... 1911_2D80_2865_0001_001B_448B... Healthy              Online                  238.47 GB GPT
```

```
PS C:\Users\hp> Get-Partition


   DiskPath : \\?\scsi#disk&ven_hgst&prod_hts725050a7e630#4&37d560ba&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            17408                                    15.98 MB Reserved
2                D           16777216                                275.97 GB Basic
3                F           296335966208                              29.3 GB Basic
4                            327793246208                               513 MB System
5                            328331165696                             159.98 GB Unknown


   DiskPath : \\?\scsi#disk&ven_nvme&prod_wdc_pc_sn520_sda#5&346797cc&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    100 MB System
2                            105906176                                   16 MB Reserved
3                C           122683392                                237.16 GB Basic
4                            254770413568                               1.2 GB Recovery
```

```
PS C:\Users\hp> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
F           Nouveau nom  NTFS           Fixed     Healthy      OK                     28.3 GB   29.3 GB
                         NTFS           Fixed     Healthy      OK                   108.29 MB    1.2 GB
C                        NTFS           Fixed     Healthy      OK                    12.18 GB 237.16 GB
D           winruth      NTFS           Fixed     Healthy      OK                    24.81 GB 275.97 GB
```

A3. Map evidence sources to system layers

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Opened files, searched terms, typed usernames, typed commands	Recent files, browser history, PowerShell or shell history	It links a user action to activity on the system
Application layer	Office document metadata (author, dates), browser cache, chat logs	Application folders (AppData) and application databases	It shows which software created or used the evidence.
OS layer	Event logs, installed programs, user profiles, account logins	Windows Event Viewer, registry, Windows logs (Linux: /var/log	It shows system events and account activity.
Disk layer	Deleted files, file fragments, partitions	File system (for example the NTFS MFT) and unallocated space	It supports data recovery and timeline analysis.
Memory/process layer	Running malware, PIDs, active network connections	RAM capture, process list (tasklist, Task Manager)	It shows the live behavior of the system.

Why evidence may exist in more than one layer

A user action goes through several layers of the computer before it is completed. Because of this, the same action can leave traces in different places.

For example, when a user opens a Word document, the system keeps a trace in many layers. The file appears in the recent files list (user layer). The document stores metadata such as the author and the last modified date (application layer). The operating system records events in logs and in the registry (OS layer). The file is read from the disk and may leave traces in the file system (disk layer). The process WINWORD.EXE is visible in memory while the program is running (memory layer).

This is important in digital forensics for two reasons:

- Corroboration: if the same evidence is found in several independent layers, it is stronger and more reliable.
- Recovery: if a user deletes a trace in one layer, for example the browser history, a copy may still exist in another layer, such as the browser cache, the unallocated disk space, or the RAM.

Part B - Program, Process and Thread Investigation

B1. Observe running processes

- Open Task Manager on Windows, or use Command Prompt/PowerShell.

- Capture a screenshot showing running processes

```
C:\Users\hp>tasklist

Nom de l'image                PID Nom de la sessio Numéro de s Utilisation
========================= ======== ================ =========== ============
System Idle Process             0 Services                   0          8 Ko
System                          4 Services                   0      3 288 Ko
Secure System                 140 Services                   0     63 244 Ko
Registry                      184 Services                   0     47 972 Ko
smss.exe                      728 Services                   0      1 552 Ko
csrss.exe                     876 Services                   0      6 092 Ko
wininit.exe                  1048 Services                   0      9 132 Ko
csrss.exe                    1056 Console                    1      8 020 Ko
winlogon.exe                 1116 Console                    1     17 088 Ko
services.exe                 1188 Services                   0     20 260 Ko
LsaIso.exe                   1208 Services                   0      4 036 Ko
lsass.exe                    1216 Services                   0     36 336 Ko
svchost.exe                  1356 Services                   0     54 364 Ko
fontdrvhost.exe              1436 Console                    1     18 100 Ko
fontdrvhost.exe              1444 Services                   0      4 324 Ko
svchost.exe                  1520 Services                   0     27 252 Ko
svchost.exe                  1572 Services                   0     11 276 Ko
WUDFHost.exe                 1624 Services                   0     21 264 Ko
svchost.exe                  1764 Services                   0      6 780 Ko
svchost.exe                  1784 Services                   0     10 348 Ko
svchost.exe                  1792 Services                   0     12 856 Ko
svchost.exe                  1800 Services                   0      6 620 Ko
dwm.exe                      1940 Console                    1    152 896 Ko
svchost.exe                  1960 Services                   0     12 272 Ko
svchost.exe                  1992 Services                   0     13 268 Ko
```

- Three processes and record their process names, PID values, and likely purpose.

Names	PID values	Likely purpose
Tasklist.exe	13900	A Windows command-line tool that lists the processes running on the computer. It appears in the list because it was running when I executed the tasklist command.
Powershell.exe	22660	The Windows PowerShell command-line shell. It lets the user run commands and scripts to manage the system and collect information.

WindowsTerminal.exe	18668	The Windows Terminal application. It is the window that hosts command-line shells such as PowerShell and Command Prompt.

B2. Explain the difference

- In your report, explain the difference between a program, a process, and a thread in your own words.

Program: is a set of instructions stored on the disk.

Process: A process is a program that is running in memory.

Thread: A thread is the smallest unit of work inside a process.

- Explain why process information may be useful during malware analysis or live forensic triage.

Process information is useful in malware analysis and live forensic triage because it shows what the computer is doing at the moment of the investigation. Malware must run as a process to work, so the process list is often the first place where an investigator can detect it. By checking the process names, PIDs, parent processes, file paths and network connections, the investigator can spot suspicious programs, for example a process with a strange name or an unusual parent such as Word starting PowerShell.

Part C - Disk Sector and Disk Size Calculation

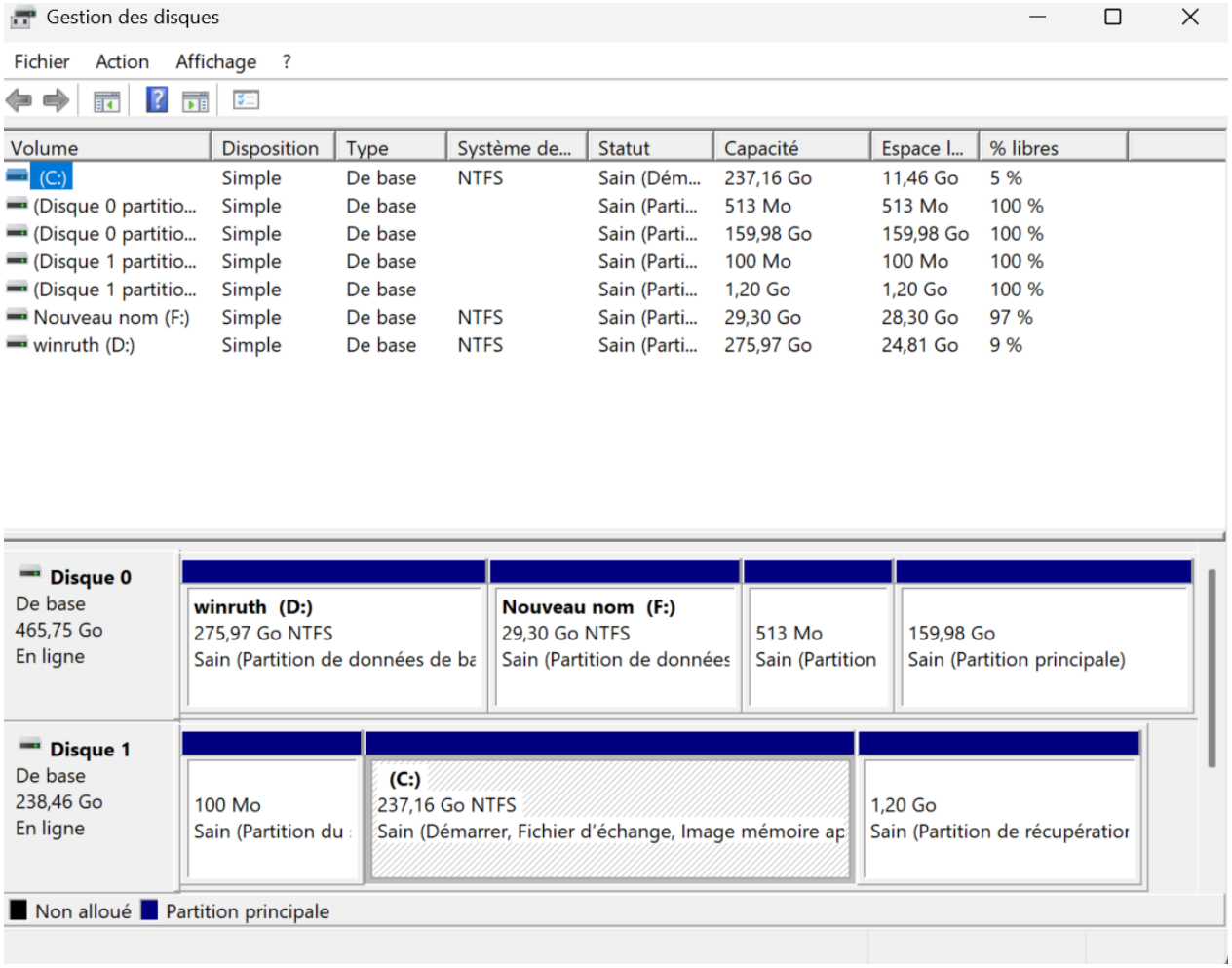
C1. Reproduce the slide calculation

sectors per track = 400, heads = 12, cylinders = 17000, and bytes per sector = 512.

Item	Formula	Answer
Total sectors	400 x 12 x 17000	81,600,000 sectors
Total bytes	81,600,000 x 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 / 1024	40,800,000 KB
Megabytes	40,800,000 / 1024	39,843.75 MB
Gigabytes	39,843.75 / 1024	38.91 GB

C2. Calculate your own disk or partition size

- Use your partition table or disk details to identify at least two partitions



Partition	Name	File system	Size
Partition 1	Winruth(D)	NTFS	275,97
Partition2	Nouveau nom (F)	NTFS	29,30 GO

• For each partition, record the start sector, number of sectors, sector size, and file system where available.

Partition	Start sector	Number of sector	Sector size	File system
D	Offset/LogicalSectorSize = 16777216/512 = 32768	Size/LogicalSectorSize = 296319188992/512= 578748416	512	NTFS
F	Offset/LogicalSectoSize = 296335966208/512 = 578781184	Size/LogicalSectorSize = 31456231424/512 = 61437952	512	NTFS

```
PS C:\Users\hp> Get-Partition | Select-Object DiskNumber, PartitionNumber, DriveLetter, Offset, Size


DiskNumber      : 0
PartitionNumber : 1
DriveLetter     :
Offset          : 17408
Size            : 16759808

DiskNumber      : 0
PartitionNumber : 2
DriveLetter     : D
Offset          : 16777216
Size            : 296319188992

DiskNumber      : 0
PartitionNumber : 3
DriveLetter     : F
Offset          : 296335966208
Size            : 31456231424

PS C:\Users\hp> Get-Disk | Select-Object Number, PartitionStyle, LogicalSectorSize, PhysicalSectorSize

Number PartitionStyle LogicalSectorSize PhysicalSectorSize
------ -------------- ----------------- ------------------
     0 GPT                          512               4096
     1 GPT                          512               4096


PS C:\Users\hp> Get-Volume | Select-Object DriveLetter, FileSystem, Size

DriveLetter FileSystem         Size
----------- ----------         ----
          F NTFS        31456227328
            NTFS          1287647232
          C NTFS       254647726080
          D NTFS       296319184896
```

• Calculate the size of each partition in GB using the formula above.

Partition D:

- Sector count = 296319188992 / 512 = 578748416 sectors
- Size in bytes = 578748416 × 512 = 296319188992 bytes
- Size in KB = 296319188992 / 1024 = 289374208 KB
- Size in MB = 289374208 / 1024 = 282592 MB
- Size in GB = 282592 / 1024 = 275.96875 GB

Partition F:

Size in GB = 61437952× 512 / 1024 / 1024 / 1024 = 31456231424 / 1073741824 = 29.2958984 GB

Part D - Safe Partition Table Inspection

D1. Windows method - Disk Management, PowerShell and DiskPart

```
PS C:\Users\hp> Get-Disk

Number Friendly Name Serial Number                    HealthStatus         OperationalStatus      Total Size Partition
                                                                                                  Style
------ ------------- -------------                    ------------         -----------------      ---------- ----------
0      HGST HTS72...        RCF50ACE0YUDTM             Healthy              Online                  465.76 GB GPT
1      WDC PC SN5... 1911_2D80_2865_0001_001B_448B... Healthy              Online                  238.47 GB GPT

PS C:\Users\hp> Get-Partition


   DiskPath : \\?\scsi#disk&ven_hgst&prod_hts725050a7e630#4&37d560ba&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            17408                                    15.98 MB Reserved
2                D           16777216                                275.97 GB Basic
3                F           296335966208                              29.3 GB Basic
4                            327793246208                              513 MB System
5                            328331165696                           159.98 GB Unknown


   DiskPath : \\?\scsi#disk&ven_nvme&prod_wdc_pc_sn520_sda#5&346797cc&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                   100 MB System
2                            105906176                                  16 MB Reserved
3                C           122683392                              237.16 GB Basic
4                            254770413568                              1.2 GB Recovery


PS C:\Users\hp> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
F           Nouveau nom  NTFS           Fixed     Healthy      OK                     28.3 GB   29.3 GB
                         NTFS           Fixed     Healthy      OK                   108.29 MB    1.2 GB
C                        NTFS           Fixed     Healthy      OK                    13.44 GB 237.16 GB
D           winruth      NTFS           Fixed     Healthy      OK                    24.81 GB 275.97 GB
```

Figure 1Powershell

```
DISKPART> list disk

  N° disque  Statut         Taille   Libre    Dyn  GPT
  ---------  -------------  -------  -------  ---  ---
  Disque 0    En ligne        465 G octets  2048 K octets        *
  Disque 1    En ligne        238 G octets  2048 K octets        *

DISKPART> select disk 0

Le disque 0 est maintenant le disque sélectionné.

DISKPART> list partition

  N° partition   Type              Taille   Décalage
  ------------   ----------------  -------  --------
  Partition 1    Réservée           15 M     17 K
  Partition 2    Principale        275 G     16 M
  Partition 3    Principale         29 G    275 G
  Partition 4    Système           513 M    305 G
  Partition 5    Inconnu           159 G    305 G

DISKPART> exit

Quitte DiskPart...
```

Figure 2DiskPart

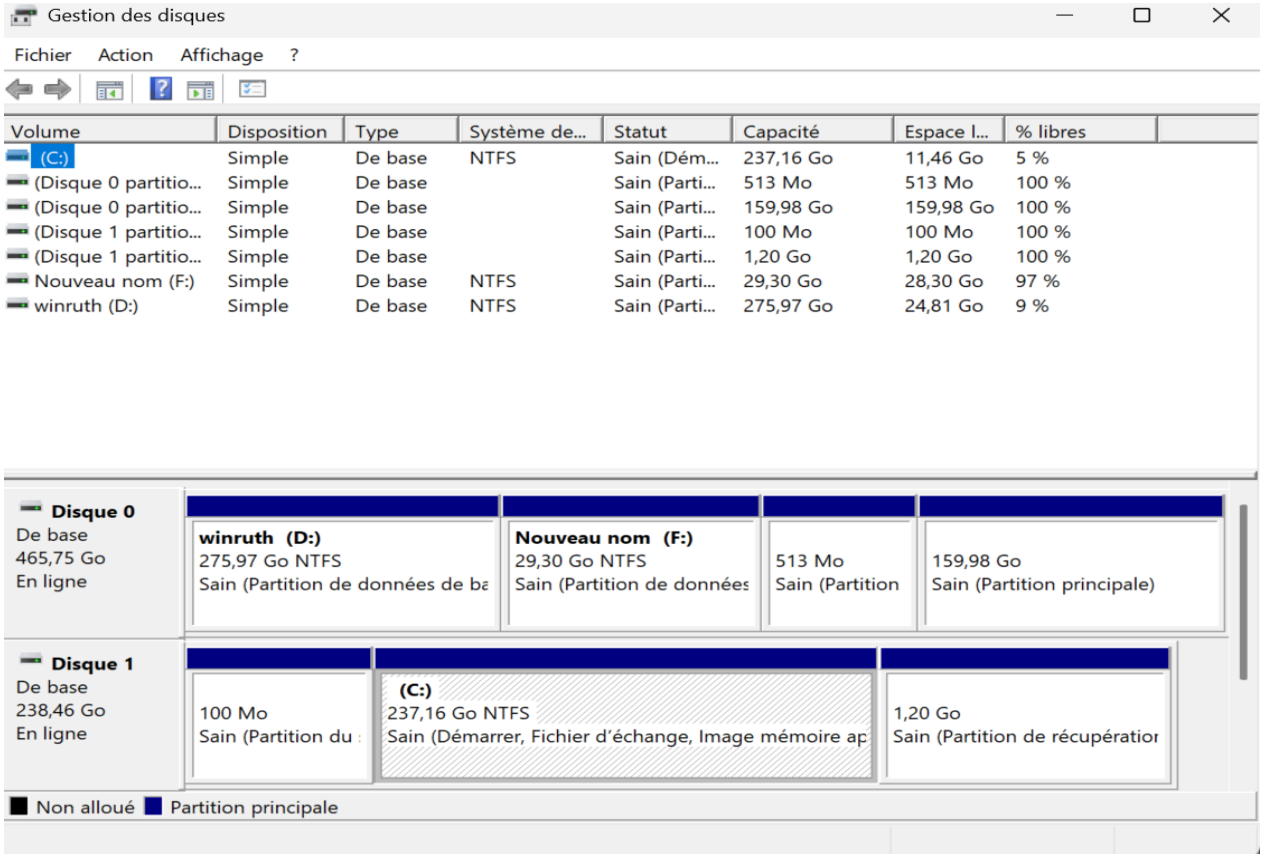


Figure 3Disk Mangement

Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

1. Power-on**:** The user presses the power button and the power supply sends electricity to the hardware.
2. POST: The firmware checks that the main hardware (CPU, RAM, keyboard, storage) is working.
3. BIOS/UEFI firmware**:** The firmware starts and initializes the devices. BIOS is the older firmware and UEFI is the modern one.
4. CMOS/firmware settings**:** The firmware reads its saved settings (boot order, boot mode, Secure Boot, date and time) from the CMOS memory, and uses them to choose the boot device.
5. MBR/GPT or EFI boot record**:** With BIOS, the firmware reads the MBR in the first sector of the disk. With UEFI, it reads the GPT and finds the boot file in the EFI System Partition.
6. Boot loader/manager**:** The boot loader (for example Windows Boot Manager, or GRUB on Linux) is loaded. It selects the operating system and loads what is needed before the kernel.

7. Kernel loading: The boot loader loads the operating system kernel and the essential drivers into memory.

8. Operating system control**:** The kernel starts the system services and the login screen. The operating system now controls the computer, and logs and user activity start to be recorded.

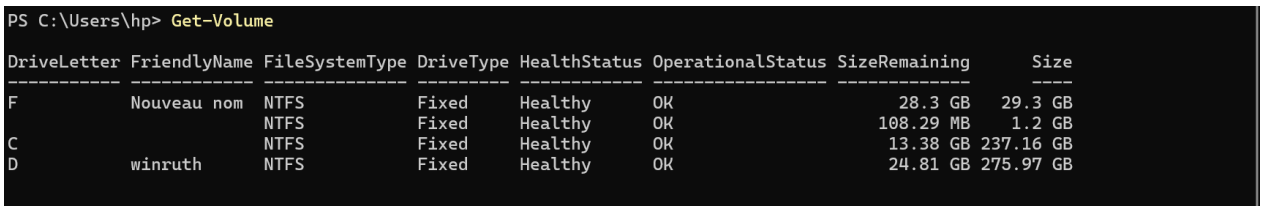


According to System Information (msinfo32), the BIOS Mode of my system is UEFI, so my computer uses UEFI firmware and not Legacy BIOS. This means that it starts from an EFI boot entry and an EFI System Partition, and not from the MBR boot code in sector 0

E2. Compare MBR and GPT

Feature	MBR	GPT
Age / design	Older partitioning system, created in the 1980s.	Modern partitioning system, used with UEFI systems.
Size limitation	Works with disks up to about 2 TB.	Supports very large disks; slide notes up to about 9.4 ZB
Partition information	The partition table and the boot code are both in the first sector (sector 0).	The partition entries are stored in a primary GPT header and copied in a backup header.
Forensic issue	If sector 0 is damaged, the partitions can become unreadable or hard to interpret.	The backup copy helps to check the table and recover the partitions.

Part F - File System Identification and Interpretation



A file system is important in digital forensics because it defines how data is stored, organized and found on a disk, so an investigator must understand it to interpret the evidence correctly. For example, NTFS, the file system of my D: and F: volumes, keeps metadata in the Master File Table (MFT), such as file names, timestamps, permissions and the location of the data. This information helps to build a timeline of user activity, to show when a file was created, modified or accessed, and to recover deleted files, because the metadata or the data of a deleted file may still exist on the disk. Knowing the file system also tells the investigator which tools and methods to use, since each file system (NTFS, FAT32, ext4) stores and deletes data differently.